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# Efficacy and safety of antazoline in the rapid cardioversion of paroxysmal atrial fibrillation (the AnPAF Study)

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**Aims** The aim of the study was to assess the clinical efficacy of antazoline, a first-generation anti-histaminic agent, in the rapid conversion of paroxysmal non-valvular atrial fibrillation (AF) to sinus rhythm in patients without heart failure.

**Methods and results** This study was a single center, randomized, double blind, placebo-controlled, superiority clinical trial. We enrolled patients with an AF episode lasting less than 43 h, in stable cardiopulmonary condition. Subjects who fulfilled the selection criteria were randomly assigned to receive intravenously either a placebo or up to 250 mg of antazoline. The primary end point was the conversion of AF to sinus rhythm confirmed in electrocardiogram (ECG). We enrolled 74 patients: 36 (48.6%) in the antazoline group and 38 (51.4%) in the control group. The mean age was 68 ± 12 years (range 31–90 years), 39 (53.3%) patients were male. The successful conversion of AF to sinus rhythm during the observation period was achieved in 26 (72.2%) patients treated with antazoline and 4 (10.5%) in the control group: RR 6.86 (95% CI: 2.66–17.72, $P < 0.0001$). Median time to conversion was 16.0 min in antazoline and 72.5 min in the control group ($P = 0.0246$). There were no cases of atrial tachycardia/flutter in the antazoline group.

**Conclusion** Intravenous antazoline was effective and safe in the rapid conversion of non-valvular paroxysmal atrial fibrillation to sinus rhythm in patients without heart failure. Clinical Trial Registration number: NCT01527279.

**Keywords** Antazoline • Atrial fibrillation • Pharmacological cardioversion • Randomized controlled trial

## Introduction

Paroxysmal atrial fibrillation (AF) is one of the most frequent problems not only in hospitalized patients but also in emergency departments.[1,2] The electrical cardioversion (CV) of AF to sinus rhythm (SR) requires sedation and for the patient to have fasted for at least 6 h. If needed, the early pharmacological cardioversion may be achieved by the administration of Class IA, IC and III anti-arrhythmic drugs: flecainide, ibutilide, dofetilide, propafenone or novel agents such as vernakalant. The effect of amiodarone, the drug with the highest efficiency, is often delayed up to a couple of hours to achieve successful cardioversion.[3–5]

Due to the potential proarrhythmic effect, lack of efficacy, long time to cardioversion or the exceptionally high cost of agents used there is still room for drugs dedicated to specific clinical situations.

Antazoline is a first-generation anti-histaminic agent with quinidine-like and anti-cholinergic properties.[6–8] The data concerning anti-arrhythmic properties of the drug are sparse. Theoretically, antazoline prolongs action potential duration and lowers its amplitude, prolongs Phase 0 duration, reduces Phase 4 of resting potential and reduces excitability of cardiac tissue. Clinically, antazoline is considered to lower the velocity of intra-atrial conduction, prolong the

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## Whats new?

- This is the first randomized controlled trial over antazoline, a first-generation anti-histaminic agent, with quinidine-like anti-arrhythmic properties used intravenously in pharmacological cardioversion of paroxysmal atrial fibrillation (AF).
- Among 74 patients: 36 in the antazoline and 38 in the placebo group, during 1.5 h observation, the successful conversion of paroxysmal AF do sinus was achieved in 26 (72.2%) and in 4 (10.5%) cases in the study and control groups, respectively: RR 6.86 (95% CI: 2.66–17.72, $P < 0.0001$).
- The median time to conversion from AF to sinus in the antazoline group was 16 min.
- There were no cases of atrial tachycardia/flutter in the antazoline group described in the literature.

atrial refractory period and improve atrioventricular conduction allowing fast ventricular response to supraventricular arrhythmias. In animal studies, the terminal elimination half-life and the mean residence time of the drug in the rat plasma were 3.53 ± 0.68 h and 3.67 ± 0.7 h, respectively.[9] In mongrel dogs, the antazoline exerted a negative effect on cardiac output and stroke volume without the influence on the blood pressure.[10] In human healthy volunteers, the terminal elimination half-life of antazoline was 2.29 h with a mean residence time 3.45 h and no data on hemodynamic effects of antazoline yet published.[11]

There have been no randomized clinical trials to evaluate the anti-arrhythmic effect of antazoline. Published studies were mainly single-arm clinical trials with no control group or a series of cases where antazoline was administered either orally or intravenously in different doses and in different arrhythmias.[12,13] The results of these studies suggested a high efficacy of antazoline in the rapid conversion of AF to SR if administered intravenously up to the total dose of 350 mg—52–83.6%.[14–16]

Antazoline has been used in clinical practice for many years due to its efficacy, safety and rapid onset of action within minutes of administration.[14,15] Antazoline can be administered intravenously in a dose of 100–300 mg during 3–10 min under strict ECG and arterial blood pressure (BP) monitoring.[12–15]

The aim of this study was to assess the clinical efficacy of antazoline in the rapid conversion of non-valvular paroxysmal AF to sinus rhythm in patients without heart failure.

# Methods

This study was a single center, randomized, double blind, placebo-controlled, superiority clinical trial. The study protocol and its amendment was in full compliance with the Declaration of Helsinki, approved by the Local Ethics Committee and registered under Clinical Trials number: NCT01527279.

The rationale and the study protocol were described previously.[17] There was a single amendment to the protocol at the beginning of enrollment: an initially intended age range from 40 to 75 years was alleviated due to the presence of patients initially excluded from the trial but who were administered antazoline outside the study.

Consecutive patients with AF lasting less than 43 h reporting to the emergency department (ED) or clinical ward of our center for cardioversion were considered for inclusion. This time frame was proposed by the Ethical Committee to allow for a cardioversion of patients who did not convert to SR during the study observation period to be performed within the 48h from the onset of AF.

The inclusion criteria were paroxysmal AF with standard indication for the cardioversion, age above 18 years, potassium blood concentration over 3.5 mmol/l, stable cardiopulmonary state defined as the absence of symptoms of acute coronary syndrome or heart failure exacerbation. The exclusion criteria comprised of: AF lasting more than 43 h, lack of written informed consent, allergy to antazoline, AF related to significant valvular disease, clinically significant heart failure or ejection fraction less than 55%, systolic blood pressure (BP) less than 100 mmHg, history of significant bradyarrhythmias without permanent pacemaker implantation, QT prolongation over 440 ms or QTc (Bazett's formula) over the population norm, heart rate more than 160', advanced liver or kidney failure, history of acute coronary syndrome, coronary artery by-pass grafting, stroke or transient ischemic attack within 30 days before enrollment, pre-excitation in the ECG, signs and symptoms of ischemia related to current episode of AF, an investigational drug used within 30 days before enrollment, pregnancy or breast feeding. The β-blockers, calcium antagonist and digoxin, were permitted for up to 2 h before study drug infusion. Treatment with intravenous anti-arrhythmic drug was not allowed for current incident of AF. Background therapy of any oral anti-arrhythmics was allowed in the study.

In order to keep the study as simple and close to clinical practice as possible, we decided to resign from an extensive patient work-up: for example, we did not perform ECHO on standard basis. This allowed for betted recruitment but obviously limited data available for analysis.

After careful consideration of inclusion and exclusion criteria, patients signed the informed consent form and were randomly assigned to the placebo or the antazoline group.

## Intervention

Apart from the assigned drug, the treatment in both groups did not differ at any time during the study. Any patient fulfilling the inclusion criteria was prepared to pharmacological CV in a standard way comprising of the standard baseline 12-lead ECG, continuous ECG monitoring, periodic non-invasive BP monitoring and intravenous line. The study drug or placebo was administered intravenously in boluses by the study nurse under supervision of the enrolling physician, both blinded to patient assignment. Drug administration was stopped in case of conversion of AF to SR, an adverse event occurrence or conversion of AF to a different supraventricular arrhythmia.

Patients assigned to the antazoline group were administered antazoline in boluses of 50 mg diluted to 10 $cm^3$ every 5 min up to a total dose of 250 mg—50 $cm^3$.

Patients assigned to the control group were administered 0.9% saline in boluses of 10 $cm^3$ every 5 min up to a maximum volume of 50 $cm^3$.

Patients' ECG and BP were continuously monitored for 1.5 h after the last dose. Further treatment of the patients including anti-coagulation depended on the clinical state and followed appropriate clinical guidelines.

## Outcomes

The primary outcome was the conversion of AF to SR confirmed in standard 12-lead ECG at the end of the observation period.

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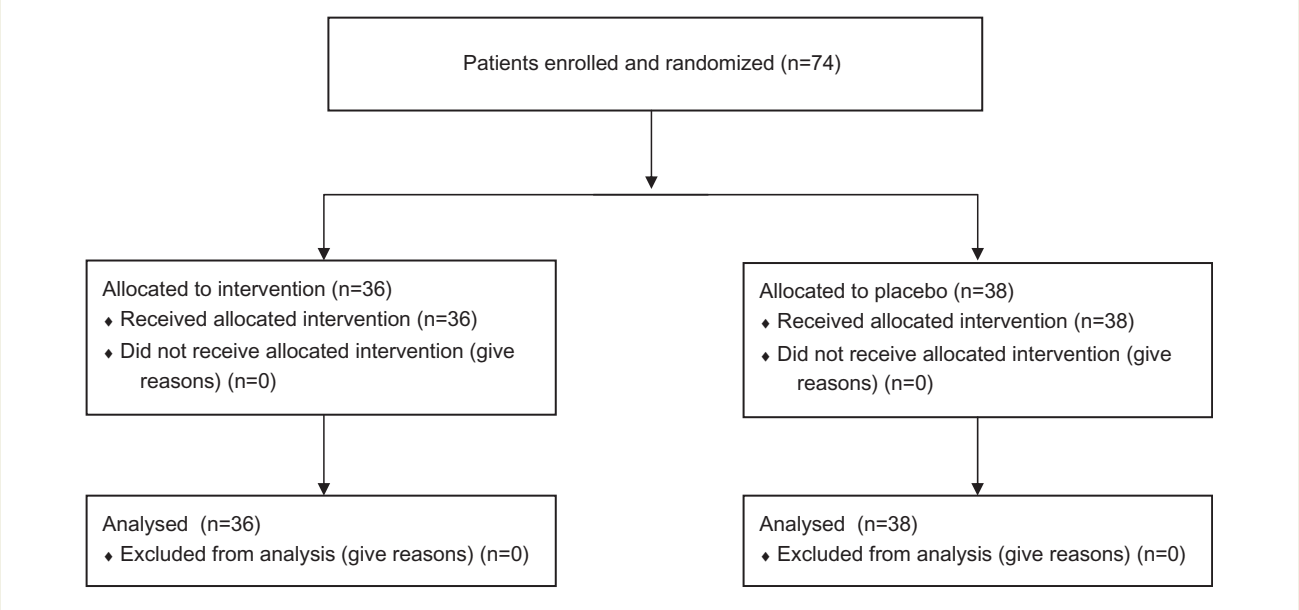


**Figure 1** Study participant CONSORT flow diagram.

Secondary end points were time to conversion, return of SR directly at the end of drug infusion, serious adverse events defined as every adverse event requiring hospitalization or prolonged observation.

Predefined safety endpoints were systolic BP less than 90 mmHg, disturbances of atrioventricular conduction, sustained supraventricular arrhythmia other than AF, new complex ventricular arrhythmia, hot flush, drowsiness, headache, nausea/vomiting, chest pain, heart rate more than 180', prolongation of QTc (Bazett's formula) in comparison with baseline.

## Statistical analysis

Based on the previously published data, we assumed a conservative 50% efficacy of antazoline and 5% efficacy of placebo in AF conversion to SR. Using the two-tailed test, a Type I error of 0.01, a power of 90% and 20% drop-out rate to fulfill the criteria of intention-to-treat analysis, we calculated 80 patients, 40 in each group, to show the superiority of antazoline over placebo.[14,17]

All analyses were conducted using SAS software (version 9.2, SAS Institute Inc., Cary, NC, USA). Normally distributed continuous data were presented in terms of mean ± standard deviation. Non-normally distributed continuous data were reported in terms of percentiles (median and interquartile range) and compared across two groups by Wilcoxon rank-sum test. Student's paired *t*- or non-parametric Mann–Whitney's test were used as appropriate to compare averages and medians between groups. A comparison between them was calculated by Pearson's $\chi^2$ test with continuity correction or Fisher's exact test. The probability of return to SR (by the time from AF) by the treatment group was graphically displayed according to the method of Kaplan–Meier, with comparison of cumulative events by the log-rank test. All hypotheses were two-tailed with a 0.05 Type I error.

The analysis was performed according to intention-to-treat principle.

The authors had full access to all data and take responsibility for the integrity of the data. All authors have read and agree to the manuscript as written.

# Results

Between November 2012 and December 2014, we enrolled 74 patients: 36 (48.6%) in the antazoline group and 38 (51.4%) in the control group (*Figure 1*). The baseline characteristics are summarized in *Table 1* and there were no statistically significant differences between study groups. The mean age was 68 ± 12 years (range 31–90 years), 39 (53.3%) patients were male.

The arrhythmia duration before drug administration was 11.2 ± 10.0 h in antazoline and 8.8 ± 8.2 h in the control group, $P$ = ns.

Successful conversion of AF to sinus rhythm during observation period was achieved in 26 (72.2%) patients treated with antazoline and in four (10.5%) in the control group: RR 6.86 (95% CI: 2.66–17.72; $P < 0.0001$), *Figure 2*. Median time to conversion was 16.0 min (9–35 min) in antazoline and 72.5 min (45–100 min) in the control group, $P = 0.0246$ (*Figure 3*).

The acute effectiveness, at the end of iv administration, was even better: 26 (72.2%) patients treated with antazoline and 3 (7.9%) in the control group: RR 9.15 (95% CI: 3.03–27.61). The mean antazoline dose in the treatment arm was 190 mg (range 50–250 mg, median 200 mg). There were no recurrences of AF after conversion to SR, both in the study drug and the control group, during the observation period.

In the antazoline group, cardioversion was successful in 3 of 4 patients on amiodarone and 7 of 10 patients on propafenone, $P = 1.00$ for both comparisons vs no conversion. In patients without any background anti-arrhythmic medications, antazoline was efficacious in 16 of 22 patients, $P = 1.00$ vs. no conversion.

The adverse effects are summarized in Table 2. There was one serious adverse event: one patient in the study group was hospitalized

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**Table 1 Characteristics of the patients at baseline**

| | Antazoline N=36 | Control N=38 | P |
|---|---|---|---|
| Age (years) | 69 ± 13 | 68 ± 12 | 0.681 |
| Heart rate, per min | 111 ± 24 | 110 ± 27 | 0.814 |
| SBP (mmHg) | 114 ±26 | 110 ± 24 | 0.300 |
| DBP (mmHg) | 88 ± 15 | 87 ± 17 | 0.421 |
| INR | 2.02 ± 1.2 | 1.99 ± 1.66 | 0.889 |
| K (mmol/dl) | 4.3 ± 0.3 | 4.3 ± 0.3 | 0.469 |
| Na (mmol/dl) | 139.2 ± 2.4 | 139.0 ± 2.4 | 0.564 |
| Concomitant diseases | | | |
| Hypertension | 52 (70.3%) | 27 (75%) | 0.386 |
| CAD | 13 (36%) | 9 (24%) | 0.368 |
| Thyroid disorders | 4 (11.1%) | 6 (15.7%) | 0.609 |
| COPD | 1 (2.8%) | 2 (5.3%) | 1.000 |
| CKD | 0 | 1 (2.6%) | 1.000 |
| Background treatment | | | |
| VKA | 29 (39.2%) | 12 (33.3%) | 0.315 |
| NOAC | 18 (24.3%) | 7 (19.4%) | 0.340 |
| Beta-blocker | 28 (77.8%) | 31 (81.6%) | 0.684 |
| Ca-blocker | 4 (11.1%) | 3 (7.9%) | 0.707 |
| Amiodarone | 4 (11.1%) | 1 (2.6%) | 0.193 |
| Propafenone | 10 (27.8%) | 18 (47.4%) | 0.082 |
| ACE-I/ARBs | 23 (63.9%) | 21 (55.3%) | 0.450 |
| Spironolactone | 13 (36.1%) | 9 (23.7%) | 0.242 |
| Diuretic | 15 (41.7%) | 16 (42.1%) | 0.969 |
| Statins | 14 (38.9%) | 15 (39.5%) | 0.958 |

SBP, systolic blood pressure; DBP, diastolic blood pressure; INR, international normalized ratio; K, potassium concentration; Na, natrium concentration; CAD, coronary artery disease; COPD, chronic obstructive pulmonary disease; CKD, chronic kidney disease; VKA, vitamin K antagonists; NOAC, new oral anti-coagulants; ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers.

due to mild dyspnoea and signs of congestion which resolved with iv diuretic. The patient recovered without any adverse consequences. Other adverse events were more frequent in the antazoline group but were transient, mild and clinically insignificant. Intravenous antazoline was well tolerated.

There were no cases of atrial tachycardia/flutter in the study group.

The mean QT/QTc intervals at the end of the observation period were 409± 43/441 ± 44 ms in the antazoline group and 362 ± 42/440 ± 31 ms in the placebo group. In patients who converted to sinus, the mean QT/QTc intervals were 422 ± 34/423 ± 27 ms in the antazoline and 399 ± 28/428 ± 17 ms in the control group.

# Discussion

Pharmacological cardioversion of paroxysmal AF is well established in clinical practice. It covers not only symptoms relief but also may reduce the length of hospital stay and hence costs of care.[1,2,18–21] In the present study, we proved the high effectiveness and rapid onset of the action of antazoline, a first-generation anti-histaminic agent with quinidine-like properties. The low risk of serious adverse events was another benefit of the investigated treatment. Moreover, antazoline has been used in clinical practice for many years in different countries due to its efficacy, safety and rapid onset of action.[14,15]

Recently published retrospective studies concerning the administration of antazoline in patients with short lasting AF during invasive electrophysiological procedures described success rates as high as 83.6–100%.[16,22]

Based on the existing data, we cannot compare antazoline with other anti-arrhythmic agents. Another drug recently indicated and registered in pharmacological cardioversion procedure was vernakalant. The clinical efficacy of vernakalant in conversion of AF to sinus rhythm is estimated at 52% and, what is more, it can be safely used in patients with coronary artery disease and in early postoperative periods after thoracic surgery.[3–5] Intravenous propafenone is relatively safe and effective. Its reported cardioversion success rates reach up to 60%.[18,19]

Intravenous amiodarone is still important and efficient especially for patients with depressed left ventricle ejection fraction. Prolonged infusion and long time to potential AF conversion to sinus rhythm limits its application in emergency settings and clinical wards.[18,19]

There was no apparent influence of the background anti-arrhythmic medication on the effectiveness of cardioversion in both groups (all comparisons, $P$ = NS) but the absolute numbers of those patients were low and the study was by far underpowered to asses any meaningful difference in this area.

Antazoline was relatively well tolerated in most patients. Adverse events were more frequent in the antazoline group but were transient, mild and without clinical significance. Nonetheless, the high proportion of any adverse events in the study group was surprising as previous observations studies did not give such high estimates.[14,15] Most frequent adverse events were hot flush, drowsiness, headache, nausea and more rarely: a bitter taste or anxiety. A single case of unmasking the heart failure symptoms responded well to intravenous diuretic and the patient was discharged 2 days after the episode without other complications. One explanation for the unexpectedly high occurrence of adverse events in the AnPAF Study was meticulous recording of all adverse events regardless of the clinical significance. Previous and recent observational studies indicated a variety of mild adverse events without providing exact estimates of rates.[14,15,23] In our recent study over antazoline administration in the emergency ward, the hospitalization rates due to adverse events were as low as 3% (10 out of 334 patients). The indications for the hospitalization were sinus bradycardia, mild hypotension, dizziness, nausea, chest discomfort or clinical suspicion of CAD and episodes of atrial flutter/tachycardia after cardioversion.[23]

In observational studies, antazoline was administered in patients with structural heart disease or even in acute myocardial infarction with good results but, generally, this should be avoided due to small numbers of patients observed.[14,15] It seems that the adverse effects of antazoline may be dose dependent and more common when the drug was administered in the dose over 250 mg.[14,15] As well as other anti-arrhythmic agents antazoline may unmask the underlying sick sinus syndrome or atrioventricular block after successful conversion to sinus rhythm.[14,15]

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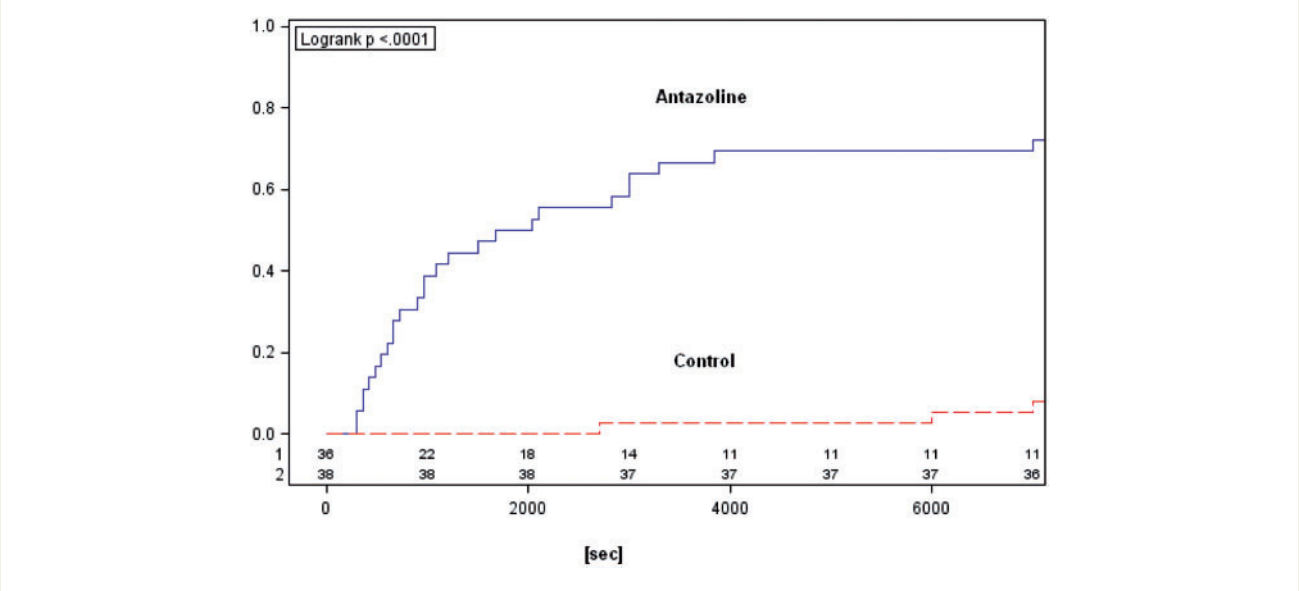


**Figure 2** Cumulative success rates based on time after the first bolus of the study drug or placebo in the control group.

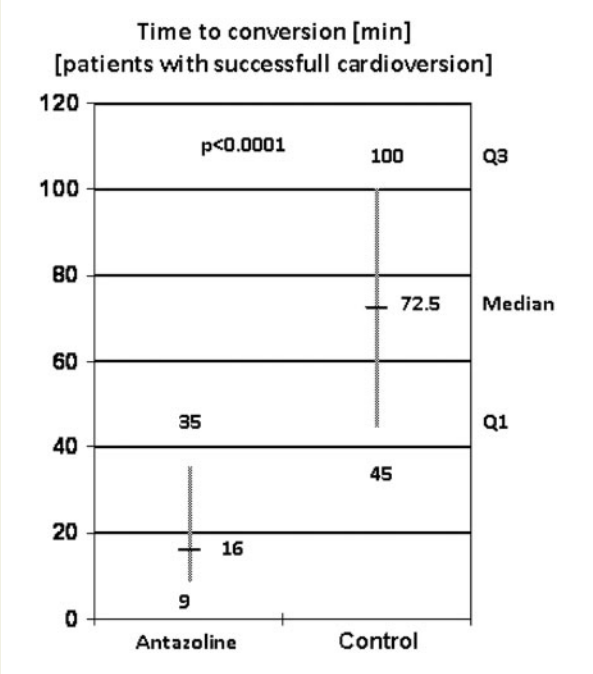


**Figure 3** Median time to conversion of atrial fibrillation to sinus rhythm based on the time after the first bolus of the study drug or placebo.

**Table 2** Adverse events

| | Antazoline *N*=36 | Control *N*=38 | *P* |
|---|---|---|---|
| Hypotension | 1 (2.8%) | 0 | <0.001* |
| Sinus tachycardia | 2 (5.6%) | 0 | |
| PVC/VT | 0 | 0 | |
| Hot Flush | 7 (19.4%) | 2 (5.3%) | |
| Drowsiness | 3 (8.3%) | 1 (2.6%) | |
| Headache | 2 (5.6%) | 0 | |
| Nausea | 2 (5.6%) | 0 | |
| Dyspnoe/chest pain | 0 | 0 | |
| QTc prolongation | 0 | 0 | |
| Tachycardia >180/min | 0 | 0 | |
| HF exacerbation | 1 (2.8%) | 0 | |
| Bradycardia | 2 (5.6%) | 0 | |
| Bitter/metallic taste | 1 (2.8%) | 0 | |
| Anxiety | 1 (2.8%) | 0 | |

SVT, supraventricular tachycardia; PVC, premature ventricular contraction; VT, ventricular tachycardia; HF, heart failure.
*P value for difference between study groups in total adverse events.

## Limitations

The two main limitations of the study are the relatively modest sample size and single center design of the trial. It should be remembered that due to the high effectiveness of the drug the calculated sample size was even lower than achieved and we deliberately increased it by about 20% to be sure to keep the intention-to-treat principle. Still, the sample size significantly limits the amount of available data on safety of the drug. The study funding did not allow for a multicenter trial.

## Conclusion

Intravenous antazoline was effective in our patients and might be safe in the rapid conversion of paroxysmal non-valvular atrial fibrillation to sinus rhythm in patients without heart failure.

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